Department of Mathematics

CC – 3 (First Order Differential Equation)

Question Bank (MDC)

1. State the order and degree of the following differential equations:

   (a) $\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^2 + 2y = 0$ (b) $\left(\frac{d^2y}{dx^2}\right)^3 + x^2\left(\frac{dy}{dx}\right)^4 = 4$ (c) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 1 + x$

   (d) $\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{2/3} = \frac{d^2y}{dx^2}$.

2. Form a differential equation from the given relation by eliminating the parameters:

   (a) $y = Ae^x + Be^{-x}$ (b) $y = \lambda x + xlogx$ (c) $xy = Ae^x + Be^{-x} + x^2$

   (d) $y = Asinx + Bsinx + xsinx$ (e) $y = e^{-x}(Acosx + Bsinx)$.

3. Find the differential equations of

   (a) all straight lines passing through the origin.

   (b) all circles passing through the origin and having centers on the x-axis.

   (c) the system of circles having a constant radius a and having centers on the x-axis.

4. Examine whether $(cosy + ycosx)dx + (sinx - xsiny)dy = 0$ is an exact differential equation.

5. Verify that $\frac{1}{y^2}$ is an integrating factor of the differential equation $y(1 + xy)dx - xdy = 0$.

6. Solve the differential equation (a) $ydx - xdy = xydx$ (b) $xdx + ydy + \frac{xdy - ydx}{x^2 + y^2} = 0$.

7. Examine whether the following differential equation is exact. If so, find the general solution to it $(y^3 + 3x^2y)dx + (x^3 + 3y^2x)dy = 0$.

8. Solve $cosx\frac{dy}{dx} + ysinx = 1$.

9. Solve $(1 + x^2)\frac{dy}{dx} + y = \tan^{-1}x$.

10. Solve $x\frac{dy}{dx} + sin2y = x^4cos^2y$.

11. Solve $\frac{dy}{dx} + \frac{y}{x}logy = \frac{y}{x^2}(log\ y)^2$.